

Motion Model-Based Animation of a Walking Character Rig

SUBMITTED BY

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Course **DSA0205 - COMPUTER VISION WITH OPENCV**

Motion Model and Implementation Pipeline

PERIODIC JOINT MOTION → FORWARD KINEMATICS

Walking Parameters

Periodic Motion Model

Joint Angle Generation

Forward Kinematics

Character Rig

Walking Animation

Motion Metrics

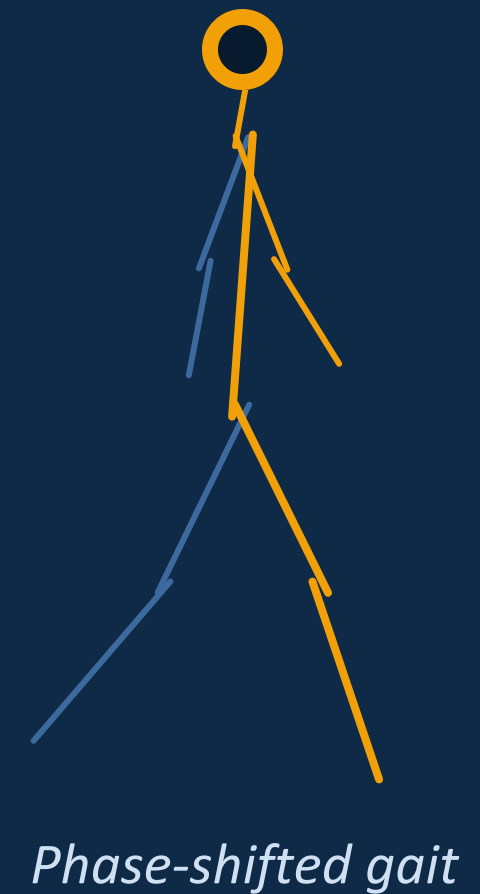
$$\theta(t) = A \cdot \sin(2\pi ft + \phi)$$

A = amplitude f = gait frequency t = time ϕ = phase

- 1 Joint angles are generated using periodic sinusoidal motion.
- 2 Left and right legs use phase-shifted motion for alternate steps.
- 3 Forward kinematics converts joint angles and segment lengths into joint positions.
- 4 Computed positions drive the animated walking character.

Objective

Generate a continuous walking animation from a mathematical motion model instead of hand-animating frames.



Walking Character Animation and Output

MATHEMATICAL CHARACTER ANIMATION — NOT MOTION CAPTURE

120

animation frames

4.00 s

duration

1.20

units/s speed

2.0

units step length

1.2 Hz

gait frequency



The model generates a continuous walking cycle by changing the joint angles over time and calculating the corresponding joint positions.

Results and Conclusion

MEASURED OUTPUT OF THE MOTION MODEL

Metric	Result
Walking speed	1.20 units/s
Total displacement	4.80 units
Animation duration	4.00 s
Estimated average speed	1.20 units/s
Left hip angle range	50.00°
Right hip angle range	50.00°
Left knee angle range	25.00°
Right knee angle range	25.00°

Interpretation

- Average speed matches the configured 1.20 units/s.
- Character travelled 4.80 units in 4 seconds.
- Both hips 50° range; both knees 25° range.
- Continuous walking animation produced.

Conclusion

The motion-model-based character rig successfully generated a repeatable walking animation using periodic joint motion and forward kinematics.

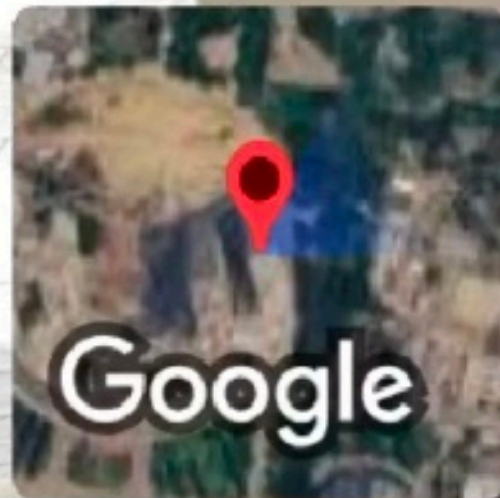
Future Scope

More realistic gait models

3D character rigs

Motion-capture validation

Detailed joint dynamics



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